SHAFIQ MAHMUD

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Portfolio | Google Scholar | ResearchGate | LinkedIn

PROFESSIONAL SUMMARY

Final-year undergraduate in Urban and Regional Planning at CUET with a strong research orientation and a technical foundation in geospatial analysis, remote sensing, and machine learning. Co-author of two preprints and a conference paper, with an undergraduate thesis in progress. Seeking graduate research opportunities.

RESEARCH INTERESTS

- **Geospatial AI (GeoAI) and Machine Learning:** spatial prediction, susceptibility modeling, model generalization, and uncertainty quantification.
- **Remote Sensing and GIS:** satellite-based environmental monitoring, geospatial data analysis, and spatial modeling.
- **Hydrology and Water Resources:** flood and drought assessment, hydrological modeling, and water-related risk analysis.
- **Climate Change, Environmental Hazards and Resilience:** agricultural drought, flood and landslide susceptibility, climate-related risks, and environmental resilience.

EDUCATION

Chittagong University of Engineering & Technology (CUET), Chattogram, Bangladesh — 2023 – Present
Bachelor of Urban and Regional Planning (BURP) | CGPA: 3.35/4.00 (up to 6th semester)

Dhaka College, Higher Secondary Certificate (HSC) | GPA: 5.00/5.00 — 2021

Basurhat A.H.C. Government High School, Secondary School Certificate (SSC) | GPA: 5.00/5.00 — 2019

RESEARCH & PUBLICATIONS

Thesis and Ongoing Research

- **Machine learning-based agricultural drought risk assessment in Bangladesh.** Undergraduate thesis, Department of Urban and Regional Planning, CUET *(ongoing)*.
- Uncertainty-informed bivariate decision zones for storm-impact susceptibility in the cyclone-exposed southwestern coast of Bangladesh *(ongoing)*.
- Coastal land dynamics of Noakhali, Bangladesh (2000–2020): a multi-sensor remote sensing analysis and engineering assessment for economic zone development *(ongoing)*.

Manuscript Under Review

1. Murad, G., Rahman, M. A., Srabony, S. C. & **Mahmud, S.** Uncertainty-aware Bayesian machine learning for landslide susceptibility mapping in the Chattogram metropolitan hill system, Bangladesh. *Journal of Mountain Science* (under review). [SCIE, Scopus, IF: 2.7, JCR Q3 | Scopus Q2] Preprint at EarthArXiv, https://doi.org/10.31223/X55J40 (2026).

Preprint

1. **Mahmud, S.**, Murad, G. & Rahman, M. A. Enhancing the generalization of flood susceptibility models: a leakage-aware ensemble framework for deltaic landscapes. Preprint at EarthArXiv, https://doi.org/10.31223/X5JJ5S (2026).

Conference Paper

1. Sarker, S., **Mahmud, S.**, Monjur, A., Talbi, M. A. & Sikder, J. National-scale drought mapping using satellite-derived biophysical indices: towards a robust drought monitoring system in Bangladesh. Presented at the *International Conference on Civil Engineering Research & Innovations*, Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh, 12–14 December 2025 (Paper ID: 336).

Technical Report

1. **Mahmud, S.**, Murad, G. & Rahman, M. A. Fragile slopes, damaged roads: an investigation into clustered landslides along the Rajasthali–Saichal segment of the Border Road, Bangladesh. Newsletter report (unpublished).

ACADEMIC PROJECTS

- **Assessment of Air Quality and Its Impact in Chattogram City Corporation.** *URP 436, Climate and Disaster Management Sessional.*
- **Flood Susceptibility Mapping in Noakhali Using Artificial Neural Networks (ANN).** *URP 366, Advanced GIS Lab.*
- **Spatio-Temporal Assessment of Nighttime Light Emissions and Urban Vulnerability: A Case Study of Dhaka.** *URP 364, Remote Sensing and Photogrammetry Lab.*

TECHNICAL SKILLS

Geospatial & Remote Sensing: ArcGIS Pro, ArcMap, QGIS, ERDAS Imagine, Google Earth Engine
Programming & Modeling: Python, R, JavaScript; machine learning for spatial prediction and susceptibility modeling
Planning & Design: AutoCAD, SketchUp | **Productivity:** Microsoft Word, Excel, PowerPoint
Media: Adobe Premiere Pro, Photoshop, Lightroom, DaVinci Resolve, Canva
Languages: Bangla (native), English

INDUSTRIAL ATTACHMENT & FIELD VISIT

- **Industrial Attachment, Dhaka North City Corporation (DNCC):** team-based study of ward-wise community facilities, benchmarked against Private Residential Land Development Rules (PRLDR) standards.
- **Field Visit, Cox's Bazar Development Authority (CoxDA) and Khurushkul Special Ashrayan Project:** observed urban management works and the world's largest climate-migrant rehabilitation initiative.

HONORS & AWARDS

- **First Prize,** "A Carbon Conscious Urban Design Idea Competition" (concept: "The Receding Membrane"), organized by UNOPS, RAJUK, and MoHPW with support from BMZ.
- **Winner (University Level),** Inter-University Quiz Competition on Spatial Planning Framework for Bangladesh, Bangladesh Institute of Planners (BIP), World Town Planning Day, November 2024.
- **First Prize,** World Town Planning Day 2025 Photography Contest (theme: Anticipatory Planning for Disaster Risk Reduction), Department of URP, CUET, November 2025.
- **Honorable Mention,** "Revealing the Unseen: Season VII" photography contest, CUET Photographic Society, June 2026.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **President,** CUET Photographic Society 2026 – 2027
- **Vice-President,** Planning Club, CUET 2026 – 2027
- **Organizing Committee Member,** TEDxCUET 2025

OUTREACH & MEDIA

- **GIS & Remote Sensing Tutorial Channel (YouTube):** creator of step-by-step tutorials on ArcGIS Pro, ArcMap, and ERDAS Imagine, covering image classification, network analysis, and ModelBuilder.
- Photography, videography, and drone-based aerial work for YPSA, RIMES, and Save the Children in Bangladesh, including landslide documentation and a flood-response documentary.